

analogue cameras is addressed by hybrid video recorders. The need for multi-location monitoring is increasing due to which video management software is in high demand.

According to Jivani, awareness regarding higher security and other benefits provided by video surveillance solutions is increasing rapidly. He also says, "Transition from analogue cameras to IP video surveillance is happening on a large scale. Today IP cameras offer higher resolution in terms of megapixels. IP cameras with 2MP and 3MP are becoming the general norm in the industry and 5MP IP cameras are also used for special applications.

"Further, these cameras also simultaneously support multiple streams with better security and networking capabilities. In addition, new IP cameras also support intelligent video analytics functions. For analogue cameras, resolution in terms of TV lines has improved from 600 TV lines to 700 TV lines and even higher."

He also indicates that there has been a significant development in the IVA technology like face recognition, people counting, missing objects and so on, which is further enhancing security.

Sehgal points out that, "Apart from technology, there has been a shift on the product side too, from box to fixed-dome cameras and pan-tilt-zoom dome cameras. In continuous pursuit of clarity, standard (medium) resolution has replaced high-resolution cameras, and from standard-dynamic-range to wide dynamic-range cameras.

"In the recording space, digital video recorders are being replaced by network video recorders and external storage. HD CCTV video surveillance equipment is the fastest-growing segment in the analogue space and the next big promising segment."

Jivani focuses on innovation in this segment. He indicates, "On the video surveillance side, the industry

needs better compression technology. A certain amount of bandwidth is required to transfer HD videos, and this could be achieved only when we have better compression techniques. The area of innovation could be video analytics, wherein, without putting a lot of time-consuming efforts, a user can directly jump to the part of the video he or she actually wants to see. There is no strong video analytic software yet that could make this possible."

Market constraints

"Business opportunity in security and surveillance is huge, but till now it has not been an attractive investment segment because of high market fragmentation, unstructured trade and commoditisation of the industry" says Sehgal.

He emphasises, "Companies and traders are selling inferior-quality products in the market, which is affecting the growth of the market and misguiding customers. The government needs to set standards for security devices that can be sold and provide guidelines for import of electronic security products in India."

Jivani supports this point as he says, "Most countries have their own standards for security products and people who install and maintain these products. It is the government's job to create standards in order to ensure a certain level of quality in products and services. Unfortunately, our country has ignored this aspect completely so far, leaving everything to manufacturers, importers, traders and customers. This has led to India becoming a dumping ground for cheap and inferior imports. Now, it is encouraging to see government talking about important initiatives such as products and skills standardisation and certification along with skill development."

Jivani emphasises, "Most countries insist on strict homologation programmes before they allow importing products. Despite India

"When it comes to enterprises, usage of cameras or security and surveillance products is not only for the purpose of security but also for productivity. When video surveillance is in place at offices, shops or any other place, productivity of the workers/employees tends to increase automatically."

—Ganesh Jivani, managing director,
Matrix Comsec



being a very large market, we have failed to establish strong standards and homologation processes."

While discussing about implementation-related challenges, Jivani says, "When we talk about implementation in video surveillance, there are a lot of bottlenecks like technological infrastructure. IP video cameras work on WAN, LAN or others. Most of the time the bandwidth required by these cameras is not available."

"Another challenge of the video surveillance market is the lack of modes of multiple site monitoring, limitations of storage and also the cost involved in the same. Multi-site management faces the challenge of the lack of proper connectivity modes, irrationally-high wiring costs and the cost of implementing a video management software for multiple locations," he adds.

He also indicates the lack of indigenous R&D and adds, "Indian industry is dependent on external R&D and technologies. Industry and government should work together to correct this by promoting indigenous R&D."

Jivani also mentions that while India is one of the largest markets in the world, there are not many manufacturers in India. India has become a virtual dumping ground for imported products. Once again, this situation is not sustainable and needs immediate correction.

Both Jivani and Sehgal further elaborate the market challenges as:

Fragmented market scenario. Indian CCTV market is still majorly